Structural Collapse of the Amutha–Perumal Scheme Based on Duo Circulant Matrices

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Abstract

Amutha and Perumal recently proposed a two-party key exchange protocol based on α-v-w-duo circulant matrices over the max-plus semiring, claiming resistance to known tropical attacks and suitability for IoT environments. This paper presents a complete cryptanalysis of this protocol. We uncover a fundamental structural weakness: the construction imposes an affine parameterization on the secret matrices, reducing each matrix to a single integer parameter. Consequently, the public messages become simple shifts of a publicly computable matrix, and the session key reduces to a constant shift of another public matrix. An eavesdropper can recover the shared secret in constant time after a one-time precomputation. The attack is deterministic, succeeds with probability 1, and requires only passive observation. We verify the attack on the authors' own example.

Keywords: Cryptanalysis; Duo circulant matrices; Key exchange; Max-plus semiring; Structural collapse; Tropical cryptography.

1 Introduction

The rapid growth of the Internet of Things (IoT) has integrated digital and physical systems across fields like manufacturing, healthcare, and smart cities [1]. Because these devices often handle sensitive information, securing their communication is critical. The foundation of this security is key exchange, which allows two parties to securely share a secret key over a public channel. This problem was first solved by Diffie and Hellman [2], whose work laid the groundwork for all modern public-key cryptography.

In 2005, Stickel proposed a key exchange protocol based on the difficulty of the discrete logarithm problem in the group of invertible matrices over a finite field [3]. However, Shpilrain later demonstrated that the protocol is insecure: an attacker can recover the secret exponents using linear algebra techniques, exploiting the invertibility of the underlying matrices [4]. This attack highlighted the need for alternative algebraic platforms that lack invertible elements and thus resist such classical cryptanalytic approaches.

Tropical cryptography emerged as a promising direction following the work of Grigoriev and Shpilrain [5], who proposed the first key exchange protocol based on tropical polynomials and the max-plus semiring. The tropical semiring $(\mathbb{R} \cup \{-\infty\}, \oplus, \odot)$, where $a \oplus b = \max(a, b)$ and $a \odot b = a + b$, offers a fundamentally different algebraic structure. Unlike classical algebras, it lacks additive inverses and, consequently, matrix invertibility—properties that had enabled earlier attacks on protocols such as Stickel's [4]. The security of tropical protocols typically relies on the difficulty of solving systems of non-linear equations over this semiring, a problem believed to be NP-hard [5]. This hardness assumption forms the basis for their claimed resistance to cryptanalytic attacks.

Despite these promising foundations, the history of tropical cryptography has been marked by a recurring pattern: elegant proposals followed by successful cryptanalysis. Kotov and Ushakov [6] developed an attack exploiting patterns in higher powers of tropical matrices, effectively breaking the original Grigoriev-Shpilrain protocol. Rudy and Monico [7] subsequently extended this line of attack with a linear periodicity approach. Grigoriev and Shpilrain proposed an enhanced protocol using semidirect products [8], but this too was broken—first by Isaac and Kahrobaei [9] and later by Jackson and Perumal [10]. A comprehensive survey by Ahmed et al. [11] catalogued the expanding landscape of tropical protocols and their corresponding vulnerabilities.

More recently, several protocols based on circulant matrices have been proposed as a means to achieve commutativity while avoiding the matrix powers that enabled previous attacks. Huang et al. [12] introduced a protocol based on upper-t-circulant matrices, while Amutha and Perumal [13] proposed protocols using lower circulant and anti-circulant matrices. However, Buchinskiy, Kotov, and Treier [14] demonstrated that these protocols are also insecure by adapting the Kotov-Ushakov attack to this setting.

The vulnerability of tropical cryptographic protocols has been further demonstrated through multiple cryptanalytic approaches. Jackson and Perumal [15] proposed an algebraic attack on key exchange schemes based on modified tropical structures, analyzing matrix sequences for almost linear periodicity properties. Jiang, Huang, and Pan [16] showed that the common secret key in Durcheva's tropical encryption scheme can be recovered by solving linear equations, accomplishing this in less than one second for matrices of order 50. Alhussaini and Sergeev [17] developed methods for solving linear systems over layered semirings, enabling key recovery in supertropical protocols. More recently, Ponmaheshkumar, Kotov, and Perumal [18] demonstrated that 3-SAT can be reduced to solving systems of quadratic polynomial equations over digital semirings, further illustrating the breadth of cryptanalytic techniques applicable to tropical and related structures.

Against this backdrop, Amutha and Perumal [19] recently proposed a two-party key exchange protocol based on α-v-w-duo circulant matrices over the max-plus semiring. The protocol introduces a commutative subset of matrices derived from w-circulant matrices with specific modifications, designed to enable efficient key generation while withstanding known tropical attacks. The authors claim that their protocol resists all prevalent attacks on tropical key exchange protocols and that its security is underpinned by the NP-hardness of solving systems of non-linear equations over the tropical semiring. They further propose integrating their protocol into IoT environments, citing its efficiency and strong security properties. A detailed description of this protocol is provided in Section 2.

In this paper, we present the first complete cryptanalysis of the Amutha-Perumal protocol [19]. We demonstrate that the construction imposes a strong affine structure on the secret matrices: every matrix in the protocol is of the form $a + C$, where a is a single secret integer and C is a fixed matrix determined solely by public parameters. This observation leads to a structural collapse of the protocol:

- The public messages satisfy $K_a = (a_1 + b_1) + M$ and $K_b = (a_2 + b_2) + M$, where M is a matrix computable from public data.
- The session key simplifies to $K = (a_1 + a_2 + b_1 + b_2) + W$, with W also publicly computable.
- An eavesdropper can recover the shared secret in constant time after a one-time precomputation of $O(m^3)$ operations.

Our attack is deterministic, succeeds with probability 1 for all parameter choices, and requires only passive observation of the exchanged messages. We verify the attack on the authors' own numerical example, recovering the exact session key.

The remainder of this paper is organized as follows. Section 2 recalls the necessary preliminaries on tropical semirings and the construction of duo circulant matrices, and provides a complete description of the target protocol. Section 3 presents our formal cryptanalysis, including the affine structure lemma, key recovery equations, complexity analysis, and verification on the authors' example. Section 4 describes our experimental results. Section 5 concludes the paper and suggests directions for future research.

2 Preliminaries

In this section we recall the necessary definitions and notations for the tropical semiring, matrix operations, and the construction of duo circulant matrices as introduced by Amutha and Perumal [19]. We then describe the two-party key exchange protocol that is the target of our cryptanalysis.

2.1 The Max-Plus Semiring

Let $\mathbb{Z}$ denote the set of integers and define $\mathbb{Z}_{\max} = \mathbb{Z} \cup \{-\infty\}$. The max-plus semiring (also called the tropical semiring) is the algebraic structure $(\mathbb{Z}_{\max}, \oplus, \odot)$ where for all $a, b \in \mathbb{Z}_{\max}$,

$$a \oplus b = \max(a, b), \qquad a \odot b = a + b,$$

with the convention that $-\infty \odot a = a \odot (-\infty) = -\infty$ for all a. The identity element for $\oplus$ is $-\infty$, and the identity element for $\odot$ is 0.

2.2 Tropical Matrix Operations

Let $M_{m\times n}(\mathbb{Z}_{\max})$ denote the set of all $m \times n$ matrices with entries in $\mathbb{Z}_{\max}$. For matrices $A = (a_{ij})$ and $B = (b_{ij})$ of appropriate dimensions, the tropical operations are defined entrywise as follows:

- Tropical addition: $(A \oplus B)_{ij} = a_{ij} \oplus b_{ij} = \max(a_{ij}, b_{ij})$.
- Tropical multiplication: If $A \in M_{m\times n}(\mathbb{Z}_{\max})$ and $B \in M_{n\times p}(\mathbb{Z}_{\max})$, then

$$(A \odot B)_{ij} = \bigoplus_{k=1}^{n} (a_{ik} \odot b_{kj}) = \max_{1\le k\le n} (a_{ik} + b_{kj}).$$

- Scalar multiplication: For $\gamma \in \mathbb{Z}_{\max}$, $(\gamma \odot A)_{ij} = \gamma + a_{ij}$.

The tropical matrix multiplication is associative and distributes over tropical addition, and the $m \times m$ identity matrix I (with zeros on the diagonal and $-\infty$ elsewhere) satisfies $I \odot A = A \odot I = A$ for any square matrix A.

2.3 Circulant and Duo Circulant Matrices

We now recall the matrix families introduced in [19].

Definition 2.1 (w-circulant matrix)**.** Let $w \in \mathbb{Z}$. A matrix $C \in M_{m\times m}(\mathbb{Z}_{\max})$ is called w-circulant if it has the form

$$C = \begin{pmatrix} r_1 & r_m & r_{m-1} & \cdots & r_2 \\ r_2 & r_1 & r_m & \cdots & r_3 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ r_m & r_{m-1} & r_{m-2} & \cdots & r_1 \end{pmatrix},$$

where the entries $r_1, r_2, \ldots, r_m \in \mathbb{Z}_{\max}$ satisfy the recurrence

$$r_k - r_{k+1} = w \qquad \text{for all } 1 \le k \le m,$$

with indices taken modulo m (i.e. $r_{m+1} = r_1$).

Definition 2.2 (α-v-w-duo circulant matrix)**.** Let $\alpha, v, w \in \mathbb{Z}$. Let $r_1, r_2, \ldots, r_m$ satisfy the recurrence of Definition 2.1 and set

$$d_1 = r_1, \qquad d_i = r_i \oplus v = r_i + v \quad (i = 2, 3, \ldots, m).$$

An α-v-w-duo circulant matrix A is then constructed by arranging the d_i in a circulant pattern and adding the constant α to most of the entries in a specific way. Concretely, the matrix is given by

$$A = \begin{pmatrix} \alpha + d_1 & \alpha + d_m & \alpha + d_{m-1} & \cdots & \alpha + d_2 \\ \alpha + d_2 & \alpha + d_1 & \alpha + d_m & \cdots & \alpha + d_3 \\ \alpha + d_3 & \alpha + d_2 & d_1 & \cdots & \alpha + d_4 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ d_m & \alpha + d_{m-1} & \alpha + d_{m-2} & \cdots & \alpha + d_1 \end{pmatrix}.$$

The set of all α-v-w-duo circulant matrices is denoted by $\mathcal{A}_v$. Replacing v by another integer c yields the family $\mathcal{B}_c$ of α-c-w-duo circulant matrices.

The security analysis in [19] establishes the following commutativity properties, which are essential for the protocol's correctness and will be used in our cryptanalysis.

Proposition 2.3 (Intra-class commutativity)**.** *For any matrices $A, A' \in \mathcal{A}_v$ we have $A \odot A' = A' \odot A$. Similarly, for any $B, B' \in \mathcal{B}_c$ we have $B \odot B' = B' \odot B$.*

2.4 The Amutha-Perumal Key Exchange Protocol [19]

Let the public parameters be chosen and published:

- an integer $m \geq 2$ (matrix dimension);
- integers $\alpha, w, v, c \in \mathbb{Z}$;
- a fixed public matrix $X \in M_{m\times m}(\mathbb{Z}_{\max})$.

Two parties, which we call Device 1 and Device 2, perform the following steps.

Key Generation (Performed Independently by Each Device)

- **Device 1:**
  1. Choose two w-circulant matrices W_1, W_2 (equivalently, choose their first entries $a_1, b_1 \in \mathbb{Z}$).
  2. Construct $A_1 \in \mathcal{A}_v$ from W_1 (using the rule with parameter v).
  3. Construct $B_1 \in \mathcal{B}_c$ from W_2 (using the rule with parameter c).
- **Device 2:**
  1. Choose two w-circulant matrices W_3, W_4 with first entries $a_2, b_2 \in \mathbb{Z}$.
  2. Construct $A_2 \in \mathcal{A}_v$ from W_3.
  3. Construct $B_2 \in \mathcal{B}_c$ from W_4.

The secret keys of Device 1 are A_1, B_1; those of Device 2 are A_2, B_2. The integers a_1, b_1, a_2, b_2 are the only secret parameters.

Message Exchange

1. Device 1 computes $K_a = A_1 \odot X \odot B_1$ and sends K_a to Device 2.
2. Device 2 computes $K_b = A_2 \odot X \odot B_2$ and sends K_b to Device 1.

Session Key Computation

- Device 1 receives K_b and computes $Y_1 = A_1 \odot K_b \odot B_1$.
- Device 2 receives K_a and computes $Y_2 = A_2 \odot K_a \odot B_2$.

The protocol claims that $Y_1 = Y_2$ and that this common value is the shared secret key K.

Correctness

From Proposition 2.3, we have $A_1 \odot A_2 = A_2 \odot A_1$ and $B_1 \odot B_2 = B_2 \odot B_1$. Therefore,

$$\begin{aligned} A_1 \odot K_b \odot B_1 &= A_1 \odot (A_2 \odot X \odot B_2) \odot B_1 \\ &= (A_1 \odot A_2) \odot X \odot (B_2 \odot B_1) \\ &= (A_2 \odot A_1) \odot X \odot (B_1 \odot B_2) \\ &= A_2 \odot (A_1 \odot X \odot B_1) \odot B_2 = A_2 \odot K_a \odot B_2. \end{aligned}$$

3 Cryptanalysis of the Amutha-Perumal Protocol

We now present a complete cryptanalysis of the two-party key exchange protocol based on α-v-w-duo circulant matrices introduced in [19]. Our analysis reveals a fundamental structural weakness: every secret matrix is an affine function of a single integer parameter. This observation reduces the protocol to a linear system, allowing an eavesdropper to recover the session key by simple arithmetic after a one-time precomputation.

3.1 Affine Structure of Duo Circulant Matrices

Recall from Definition 2.2 that a matrix $A \in \mathcal{A}_v$ is constructed from a sequence $p_1, \ldots, p_m$ satisfying the public recurrence $p_k - p_{k+1} = w$. Hence

$$p_k = p_1 - (k-1)w \qquad (k = 1, \ldots, m).$$

All entries of A are linear combinations of the p_k with known coefficients. For example, the first row is

$$\big(\alpha + p_1,\ \alpha + p_m + v,\ \alpha + p_{m-1} + v,\ \ldots,\ p_2\big),$$

and similar expressions hold for every other row. Substituting the formula for p_k shows that each entry is of the form p_1 plus a constant that depends only on the public parameters (α, w, v) and the position (i, j). Consequently, there exists a fixed matrix $C^{(A)} \in \mathbb{Z}_{\max}^{m \times m}$ (depending only on public data) such that

$$A = a + C^{(A)},$$

where $a = p_1$ is the unique secret integer defining A, and addition is entrywise. The same reasoning applied to $\mathcal{B}_c$ yields a fixed matrix $C^{(B)}$ with

$$B = b + C^{(B)},$$

for a unique integer b.

Lemma 3.1 (Affine parameterisation). *There exists a fixed matrix $C^{(A)} \in \mathbb{Z}_{\max}^{m\times m}$, depending solely on the public parameters α, w, v, m, such that every matrix $A \in \mathcal{A}_v$ can be expressed as $A = a + C^{(A)}$ where $a = p_1$ is the unique secret integer defining A. Similarly, for the family $\mathcal{B}_c$ there exists a fixed matrix $C^{(B)}$ such that every $B \in \mathcal{B}_c$ satisfies $B = b + C^{(B)}$ with secret integer b.*

Proof. From the construction rules, each entry of A is a linear expression in $p_1, \ldots, p_m$ with known coefficients. Using the recurrence (3.1), every entry becomes p_1 plus a constant that depends only on the public parameters and the position. Collecting these constants entrywise yields $C^{(A)}$. For example, for $m = 4$ the constant matrix is

$$C^{(A)} = \begin{pmatrix} \alpha & \alpha - 3w + v & \alpha - 2w + v & -w \\ \alpha - w & \alpha & \alpha - 3w + v & \alpha - 2w + v \\ \alpha - 2w & \alpha - w & \alpha & \alpha - 3w + v \\ -3w & \alpha - 2w & \alpha - w & \alpha \end{pmatrix},$$

and the general form follows the same pattern. The same reasoning applied to $\mathcal{B}_c$ gives $C^{(B)}$. □

Thus the entire secret information of the two parties consists of four integers:

$$a_1, b_1 \text{ (Device 1)}, \qquad a_2, b_2 \text{ (Device 2)},$$

with $A_i = a_i + C^{(A)},\ B_i = b_i + C^{(B)}$ $(i = 1, 2)$.

3.2 The Public Messages

Using the affine representation, the first public message becomes

$$K_a = (a_1 + C^{(A)}) \odot X \odot (b_1 + C^{(B)}).$$

For each entry (i, j),

$$\begin{aligned} K_a(i,j) &= \max_{k,\ell}\big(a_1 + C^{(A)}_{ik} + X(k,\ell) + b_1 + C^{(B)}_{\ell j}\big) \\ &= a_1 + b_1 + \max_{k,\ell}\big(C^{(A)}_{ik} + X(k,\ell) + C^{(B)}_{\ell j}\big). \end{aligned}$$

Define the matrix M by

$$M_{ij} = \max_{k,\ell}\big(C^{(A)}_{ik} + X(k,\ell) + C^{(B)}_{\ell j}\big).$$

Then (3.2) can be written compactly as

$$K_a = (a_1 + b_1) + M,$$

where the addition is entrywise. Analogously,

$$K_b = (a_2 + b_2) + M.$$

Crucially, the matrices $C^{(A)}, C^{(B)}$ are public (they are obtained by setting the secret parameter to zero in the construction rules), and therefore M is publicly computable.

3.3 Recovery of the Session Key

The session key is $K = A_1 \odot K_b \odot B_1$. Substituting the affine forms and using (3.2):

$$\begin{aligned} K &= (a_1 + C^{(A)}) \odot \big((a_2 + b_2) + M\big) \odot (b_1 + C^{(B)}) \\ &= (a_1 + C^{(A)}) \odot (a_2 + C^{(A)}) \odot X \odot (b_2 + C^{(B)}) \odot (b_1 + C^{(B)}), \end{aligned}$$

where we used $(a_2 + b_2) + M = (a_2 + C^{(A)}) \odot X \odot (b_2 + C^{(B)})$ from the definition of M and (3.2).

Because matrices in $\mathcal{A}_v$ commute (Proposition 2.3), we have

$$(a_1 + C^{(A)}) \odot (a_2 + C^{(A)}) = (a_1 + a_2) + (C^{(A)} \odot C^{(A)}).$$

Similarly,

$$(b_1 + C^{(B)}) \odot (b_2 + C^{(B)}) = (b_1 + b_2) + (C^{(B)} \odot C^{(B)}).$$

Define the precomputable matrices

$$U = C^{(A)} \odot C^{(A)}, \quad V = C^{(B)} \odot C^{(B)}, \quad W = U \odot X \odot V.$$

Then

$$K = \big((a_1 + a_2) + U\big) \odot X \odot \big((b_1 + b_2) + V\big).$$

For any scalars s, t and fixed matrices U, V, we have the identity

$$(s + U) \odot X \odot (t + V) = s + t + (U \odot X \odot V),$$

because tropical multiplication distributes and adding a scalar simply shifts all entries. Applying this with $s = a_1 + a_2$ and $t = b_1 + b_2$ yields

$$K = (a_1 + a_2 + b_1 + b_2) + W.$$

Thus the session key is a constant shift of the publicly computable matrix W by the sum $S = a_1 + a_2 + b_1 + b_2$.

From (3.2) we obtain the individual sums directly:

$$a_1 + b_1 = K_a(1,1) - M_{11}, \qquad a_2 + b_2 = K_b(1,1) - M_{11}.$$

(Any entry (i, j) may be used; consistency is guaranteed by the protocol.) Letting $s = a_1 + b_1$ and $t = a_2 + b_2$, we have $S = s + t$. Therefore the attacker computes

$$K = (s + t) + W.$$

3.4 Complexity

The matrices $C^{(A)}$ and $C^{(B)}$ are obtained directly from the public construction rules without any computation. The tropical products $M = C^{(A)} \odot X \odot C^{(B)}$, $U = C^{(A)} \odot C^{(A)}$, $V = C^{(B)} \odot C^{(B)}$, and $W = U \odot X \odot V$ each require $O(m^3)$ time using standard algorithms (or $O(m^2)$ if the circulant structure is exploited). These computations are performed **once** per parameter set, independently of the number of sessions.

Per session, the attacker only needs to read two entries from the intercepted messages K_a and K_b, perform two subtractions and one addition — **constant time**.

Theorem 3.2. *The Amutha-Perumal key exchange protocol is completely broken. An eavesdropper who observes the public messages K_a and K_b can recover the exact session key K in $O(m^3)$ precomputation time and $O(1)$ online time. The attack is deterministic and succeeds with probability 1.*

3.5 Verification on the Authors' Example

We apply the attack to the numerical example given in [19, Section 3.1]. The public parameters are $m = 4,\ \alpha = 924,\ v = 15,\ c = -221,\ w = 8$. From the construction rules we obtain the constant matrices $C^{(A)}$ and $C^{(B)}$ (they coincide with the matrices labelled $\mathcal{A}$ and $\mathcal{B}$ in the paper after subtracting the respective first-row parameters). Computing $M = C^{(A)} \odot X \odot C^{(B)}$ yields the entry

$$M_{11} = 113026,$$

which can be verified directly by evaluating

$$M_{11} = \max_{k,\ell}\big(C^{(A)}_{1k} + X(k,\ell) + C^{(B)}_{\ell 1}\big)$$

using the published matrices.

From the published K_a and K_b we have

$$K_a(1,1) = 113198, \qquad K_b(1,1) = 112886.$$

Hence

$$s = 113198 - 113026 = 172, \qquad t = 112886 - 113026 = -140,$$

so $S = 32$.

Next compute $U = C^{(A)} \odot C^{(A)}$, $V = C^{(B)} \odot C^{(B)}$, and $W = U \odot X \odot V$. Then

$$K = S + W = \begin{bmatrix} 115622 & 115838 & 115846 & 116062 \\ 115629 & 115845 & 115853 & 116069 \\ 115623 & 115839 & 115847 & 116063 \\ 115630 & 115846 & 115854 & 116070 \end{bmatrix},$$

which exactly matches the published shared key Y.

4 Experimental Results

We implemented the proposed structural collapse attack in Python to validate its effectiveness and efficiency.

All experiments were conducted on a system with an 11th Gen Intel(R) Core(TM) i5-11300H processor running at 3.10 GHz, equipped with 8 GB of RAM and operating on Windows 11 Pro 64-bit. The attack was implemented in Python 3.10.

Following the methodology of Buchinskiy et al. [14], we performed experiments for matrix dimensions $m = 5, 10, 25, 50$. For each dimension, we generated 100 random instances with

Algorithm 1 Structural Collapse Attack on the Amutha-Perumal Protocol

Require: Public parameters m, α, w, v, c, public matrix X, intercepted messages K_a, K_b
Ensure: Shared secret key K **Precomputation phase (one-time)**

Construct constant matrices $C^{(A)}$ and $C^{(B)}$ by setting $p_1 = 0$ in the recurrence (3.1) (see Lemma 3.1)
Compute $M \leftarrow C^{(A)} \odot X \odot C^{(B)}$ ▷ see (3.2)
Compute $U \leftarrow C^{(A)} \odot C^{(A)}$
Compute $V \leftarrow C^{(B)} \odot C^{(B)}$
Compute $W \leftarrow U \odot X \odot V$ ▷ see (3.3) **Online phase (per session)**
Extract $K_a(1,1)$ and $K_b(1,1)$ from intercepted messages
$s \leftarrow K_a(1,1) - M_{11}$ ▷ see (3.3)
$t \leftarrow K_b(1,1) - M_{11}$
$K \leftarrow (s+t) + W$ ▷ Entrywise addition of scalar $s+t$ to matrix W; see (3.3)
return K

Table 1: Average running time of the structural collapse attack (in seconds)

Matrix size m	**Precomputation time (s)**	**Online time (s)**	**Total time (s)**
5	0.00009	0.00000	0.00010
10	0.00058	0.00001	0.00059
25	0.00939	0.00005	0.00944
50	0.07130	0.00019	0.07149

parameters chosen uniformly from the range $[-10^5, 10^5]$, including matrix entries, public parameters α, w, v, c, and secret parameters a_1, b_1, a_2, b_2.

Table 1 presents the average running times for both precomputation and online phases across all 100 random instances for each matrix dimension.

The precomputation time grows as $O(m^3)$, consistent with the complexity of tropical matrix multiplication. The online phase is constant-time (less than 0.0002 seconds) regardless of matrix dimension. For all 400 test instances (100 per matrix dimension), the attack successfully recovered the exact session key. The success rate is 100%, confirming that our deterministic attack always succeeds regardless of parameter choices. Table 2 compares our attack with the approach of Buchinskiy, Kotov, and Treier [14].

Table 2: Comparison of our attack with Buchinskiy et al. [14]

Metric	**Our Attack**	**Buchinskiy et al.**
Methodology	Direct algebraic simplification	Cover enumeration + linear programming
Nature	Deterministic	Heuristic
Success rate	100%	100% (experimentally)
Online time ($m = 50$)	0.00019 s	6.63 s (total time)
Search required?	No	Yes (enumerates minimal covers)
Identifies root cause?	Yes (affine collapse)	No

5 Conclusion and Future Work

We have presented a complete cryptanalysis of the Amutha-Perumal two-party key exchange protocol based on α-v-w-duo circulant matrices over the max-plus semiring. Our analysis reveals a fundamental structural weakness: the construction imposes an affine parameterisation on the secret matrices, reducing each matrix to a single integer parameter. This observation leads to a deterministic attack that recovers the session key in constant time after a one-time precomputation of $O(m^3)$ operations.

The attack requires only passive observation of the public messages K_a and K_b, succeeds with probability 1 for all parameter choices, and refutes every security claim made by the original authors. The protocol provides no confidentiality and is completely unsuitable for IoT or any other practical application.

The root cause of insecurity lies in the linear recurrence (3.1), which forces all matrix entries to be affine functions of a single secret value. This serves as a cautionary principle for cryptographic design: achieving commutativity through linear dependencies can inadvertently destroy all security.

Future work should focus on designing tropical key exchange protocols that are provably secure against structural attacks, avoiding linear parameterisations and instead relying on genuinely non-linear operations or well-studied hard problems in tropical algebra.

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